

Nikhil Rout

📍 Hyderabad, India ✉ nikhilrout97@gmail.com in nikhil-rout

🐙 NikhilRout 🔗 NikhilRout.github.io

Education

Bachelor of Technology, Electronics and Communication Engineering 2022 – 2026
Vellore Institute of Technology, Chennai CGPA: 9.15/10

Research Experience

Research Intern Remote
Vortex GPGPU, ORCAS Lab, UCLA May 2025 – Present

- Developed a Tensor Core Unit (TCU) extension for the open-source RISC-V based Vortex GPGPU, accelerating WMMA operations in deep learning kernels
- Architected and implemented a configurable Microscaling (MX) Fused Dot Product datapath to explore custom block-floating-point formats
- Performed microarchitecture design space explorations, evaluating format-wise shared multipliers, Modulo-4 operand grouping CSA, and threads/warp arithmetic intensity scaling
- Introduced a novel sparse lane mask clock-gating strategy for dynamic power reduction in workloads exhibiting dual-side unstructured sparsity
- Validated TCU FP numerical accuracy by porting the FTTN benchmark to Vortex WMMA API
- Achieved 4-cycle latency at 300 MHz Fmax on the Xilinx U55C FPGA, delivering $\sim 2.7\text{-}7.9\times$ speedup over the open-source baselines (Berkeley HardFloat, FPnew) at less than 60% the area cost
- Demonstrated end-to-end speedups on Llama2-inference benchmarks: 9.2x (prefill) and 2.0x (decode)

Summer Research Intern Chennai
Centre for Nanoelectronics and VLSI Design, VIT Chennai May 2024 – Jul 2024

- Investigated radix-2 DIT and DIF butterfly structures to evaluate optimal dataflow for FFT and Number Theoretic Transform hardware accelerators
- Designed pipelined NTT/INTT cores with modulo-arithmetic operations for polynomial multiplication in lattice-based post-quantum cryptography schemes on FPGAs

Research Publications

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- [1] **Nikhil Rout**, Blaise Tine, “[Ten-Four: An Open-Source Fused Dot Product Unit for Mixed-Precision GPGPU Tensor Cores](#)”. Preprint, Under Review.
- [2] **Nikhil Rout**, Jean Jenifer Nesam J, “[Optimizing RGB to Grayscale, Gaussian Blur and Sobel-Filter operations on FPGAs for reduced dynamic power consumption](#)”. In *IEEE Artificial Intelligence For Internet of Things (AIIoT)*, 2024.

Talks and Workshops

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- [Rethinking Tensor Core Microarchitecture for Emerging Microscaling Format Support on GPUs](#)
Workshop on Open-Source Computer Architecture Research (OSCAR) @ ISCA 2026
 - [Investigating Fused Dot Product Microarchitecture for the Vortex Tensor Core Unit Extension](#)
Vortex Workshop and Tutorial @ MICRO 2025

Skills and Competencies

Languages: Verilog/SystemVerilog, C/C++, OpenCL, CUDA, Python/PyTorch

Tools: Quartus Prime, Xilinx Vivado/Vitis, Synopsys VCS/DC, TCL, CMake, Git, LaTeX

Simulators: Gem5, RISC-V Spike, GPGPU-Sim, Accel-Sim, Vortex SimX, Ramulator

Honors, Awards and Service

- Student Volunteer, HotChips 2026
- Student Travel Grant, IEEE/ACM ISCA 2026 (\$1000)
- Student Travel Grant, IEEE/ACM MICRO 2025 (\$580)
- Outstanding Teaching Assistant, VIT Chennai (FPGA System Design 2025)
- Best Paper Award, IEEE AIIoT 2024

References

Dr. Blaise-Pascal Tine, *Assistant Professor, UCLA*

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Dr. Jean Jenifer Nesam J, *Assistant Professor, VIT Chennai*

✉ jeanjenifernesam.j@vit.ac.in